EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

ATTACKING REAGENT

1. Which of the following species is an electrophile
(1) RNH_2 (2) SO_3
(3) $NO_3^{\ominus}$ (4) ROH

GC0001

2. Which of the following acts as a nucleophile?

(1) $\overset{\oplus}{N}O_2$ (2) $:CCl_2$

(3) $\overset{\ominus}{N}H_2$ (4) $^{\bullet}CH_3$

GC0002

REACTION INTERMEDIATES

3. Which of the following contains only three pair of electrons :
(1) Carbanion (2) Carbocation
(3) Carbon free radical (4) None

GC0003

4. Carbanion is a :-
(1) Base
(2) Nucleophile
(3) Both the above
(4) None

GC0004

5. In which structure carbon does not act as a electrophile.
(1) $CH_3–CH_2–Cl$
(2) $CH_3–CO–CH_3$
(3) 
(4) $CH_3–CN$

GC0160

6. Wrong statement regarding methyl carbonium ion $(\overset{\oplus}{C}H_3)$.
(1) It is sp^2 hybridised
(2) Vacant orbital is sp^2 hybridised
(3) Vacant orbital is perpendicular to molecular planar and in pure p-orbital
(4) It is electrophile with sextet of electron

GC0161

I-EFFECT

7. $\overset{\ominus}{C}H_3$ is less stable than :-

(1) $CH_3—\overset{\ominus}{C}H_2$ (2) $CH_3—\overset{\ominus}{C}H—CH_3$

(3) $\overset{\ominus}{C}H_2—NO_2$ (4) $CH_3—\overset{\ominus}{C}H—C_2H_5$

GC0005

8. Decreasing order of –I effect of the triad [$–NO_2$, $–\overset{\oplus}{N}H_3$, –CN] is :-

(1) $–\overset{\oplus}{N}H_3 > –NO_2 > –CN$

(2) $–\overset{\oplus}{N}H_3 > –CN > NO_2$

(3) $–CN > –NO_2 > –\overset{\oplus}{N}H_3$

(4) $–NO_2 > –CN > –\overset{\oplus}{N}H_3$

GC0006

9. Most stable carbanion is :-

(1) $HC\equiv\overset{\ominus}{C}$ (2) $H_2C=\overset{\ominus}{C}H$

(3) $CH_3–\underset{CH_3}{\overset{CH_3}{C}}—\overset{\ominus}{C}H_2$ (4) $CH_3–\underset{CH_3}{C}=\overset{\ominus}{C}H$

GC0007

10. The correct order of stability of given carbanions will be :-

$CH_3–\overset{\ominus}{C}H_2$ (I) $CH_2=\overset{\ominus}{C}H$ (II) $HC\equiv\overset{\ominus}{C}$ (III)

(1) I > II > III (2) III > II > I
(3) I > III > II (4) II > I > III

GC0008

11. Which is most basic among the following :-
(1) CH_3NH_2 (2) $CH_3CH_2NH_2$
(3) NH_3 (4) $(CH_3)_2CHNH_2$

GC0009

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12. Which of the following has maximum pK_a :-
(1) CH_2FCOOH (2) $CH_2ClCOOH$
(3) CH_3COOH (4) HCOOH
GC0010

13. Which of the following is most acidic ?
(1) Methoxy acetic acid (2) Acetic acid
(3) Chloro acetic acid (4) Trifluoroacetic acid
GC0011

14. Which of the following show + I-effect :-
(1) – OH (2) $-OCH_3$ (3) $-CH_3$ (4) – Cl
GC0012

15. Among the following the most highly ionised in water is:
(1) $CH_3CH_2CHClCOOH$
(2) $CH_3CH_2CCl_2COOH$
(3) $CH_3CHClCH_2COOH$
(4) $CH_2ClCH_2CH_2COOH$
GC0013

16. The strongest acid amongst the following compounds is ?
(1) $CH_3CH_2CH(Cl)CO_2H$
(2) $ClCH_2CH_2CH_2COOH$
(3) CH_3COOH
(4) HCOOH
GC0014

17. Which of the following acids is stronger than acetic acid :-
(1) Propanoic acid (2) Formic acid
(3) Butyric acid (4) Iso butyric acid
GC0015

18. Which of the following acids have the lowest pK_a value :-
(1) $CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-COOH$
(2) $Cl-CH_2-CH_2-COOH$
(3) CCl_3COOH
(4) $CHCl_2COOH$
GC0016

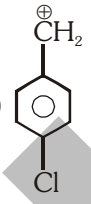
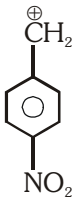
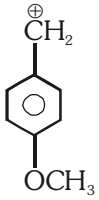
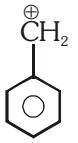
19. In which σ bond, the inductive effect is minimum ?

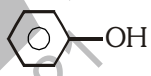
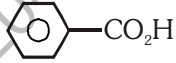
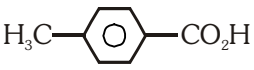
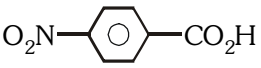
$$\underset{(d)}{C}-\underset{(c)}{C}-\underset{(b)}{C}-\underset{(a)}{C}-Cl$$

(1) a (2) b (3) c (4) d
GC0162

20. Arrange the following in the acidic strength order :
(a) NO_2CH_2COOH (b) $NC-CH_2COOH$
(c) CCl_3COOH (d) $CHCl_2COOH$
(1) a > b > c > d (2) c > d > a > b
(3) b > a > d > c (4) c > a > b > d
GC0163

R- OR M-EFFECT

21. Most stable carbocation is :-
(1)  (2)  (3)  (4) 
GC0017

22. Most acidic compound is :-
(1) 
(2) 
(3) 
(4) 
GC0018

23. Which resonating structure of vinyl chloride is least stable :-
(1) $CH_2=CH-\ddot{\underset{\cdot\cdot}{Cl}}:$
(2) $\overset{\ominus}{C}H_2-CH=\overset{\oplus}{Cl}$
(3) $\overset{\ominus}{C}H_2-\overset{\oplus}{C}H-Cl$
(4) All have equal stability
GC0019

24. The stabilization due to resonance is maximum in
(1)  (2)  (3)  (4) 
GC0020

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25. In which of the following compounds carbon–chlorine bond distance is minimum :-

(1) CH_3—Cl

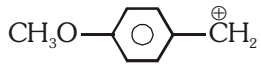
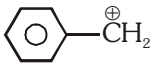
(2) C_6H_5—CH_2—Cl

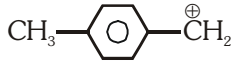
(3) CH_2=CH—Cl

(4) CH_2=CH—CH_2—Cl

GC0021

26. Consider the following carbocations

(a)  (b) 

(c)  (d) $CH_3—\overset{\oplus}{C}H_2$

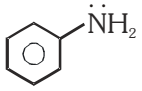
The relative stabilities of these carbocations are such that :-

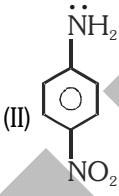
(1) d < b < c < a (2) b < d < c < a

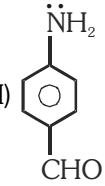
(3) d < b < a < c (4) b < d < a < c

GC0022

27. Arrange in decreasing order of basic strength :

(I) 

(II) 

(III) 

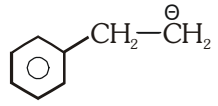
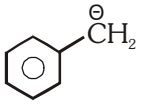
(IV) 

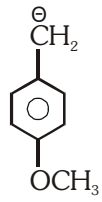
(1) I > II > III > IV (2) II > III > I > IV

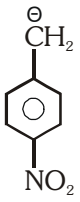
(3) IV > I > III > II (4) IV > I > II > III

GC0024

28. The most stable carbanion among the following is

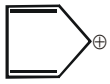
(1)  (2) 

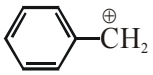
(3) 

(4) 

GC0025

29. Which of the following is most stable carbocation:-

(1)  (2) 

(3)  (4) CH_2=CH–$\overset{\oplus}{C}H_2$

GC0026

30. The oxygen atom in phenol –

(1) exhibits only inductive effect

(2) exhibits only resonance effect

(3) has more dominating resonance effect than inductive effect

(4) has more dominating inductive effect than resonance effect

GC0027

31. Which is incorrect stability order :-

(1) $CH_2 = CH - \overset{\oplus}{C}H_2 > CH_3 - \overset{\oplus}{C}H - CH_3$

(2) $CH_2 = \overset{\oplus}{C}H < CH_3 - \overset{\oplus}{C}H_2$

(3) $CH_3 - CH_2 - \overset{\ominus}{C}H_2 > CH_3 - \overset{\ominus}{C}H - CH_3$

(4) $CH_3 - \overset{\oplus}{C}H_2 > CH_3O - \overset{\oplus}{C}H_2$

GC0028

32. Mesomeric effect is due to :-

(1) Delocalization of σ e$\bar{s}$

(2) Delocalization of π e$\bar{s}$

(3) Migration of H – atom

(4) Migration of proton

GC0029

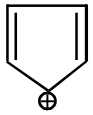
33. Among the following the pKa is minimum for :–

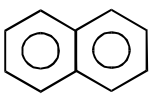
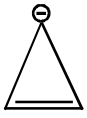
(1) C_6H_5OH (2) HCOOH

(3) C_2H_5OH (4) $CH_3C{\equiv}CH$

GC0031

34. Among the following the aromatic compound is –

(1)  (2) 

(3)  (4) 

GC0032

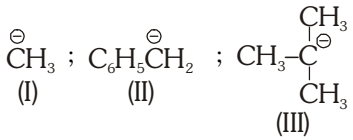
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35. Which is aromatic compound among the following

(1) 
(2) 
(3) 
(4) All the above

GC0033

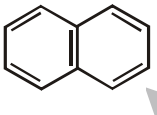
36. Select the correct option for stability of following carbanions :


(I) (II) (III)

(1) I > II > III
(2) II > I > III
(3) III > II > I
(4) II > III > I

GC0034

37. The non aromatic compound among the following is :-

(1) 
(2) 
(3) 
(4) 

GC0036

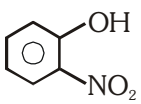
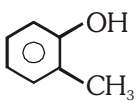
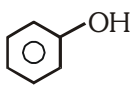
38. The correct order of acidic strength of the following compounds is :-

A. Phenol
B. p-Cresol
C. m-Nitrophenol
D. p- Nitrophenol

(1) C > B > A > D
(2) D > C > A > B
(3) B > D > A > C
(4) A > B > D > C

GC0037

39. Which one of the following compounds is most acidic:-

(1) 
(2) 
(3) 
(4) $ClCH_2CH_2OH$

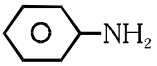
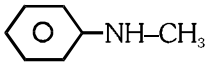
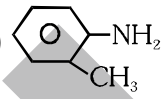
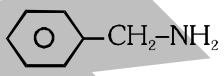
GC0038

40. Which of the following is most acidic :-

(1) phenol
(2) benzyl alcohol
(3) m-chloro phenol
(4) cyclohexanol

GC0039

41. Which of the following is the strongest base :-

(1) 
(2) 
(3) 
(4) 

GC0040

42. The least reactive chlorine is present in –

(1) Methyl chloride
(2) Allyl chloride
(3) Ethyl chloride
(4) Vinyl chloride

GC0045

43. Which one of the following resonating structures of 1–methoxy–1,3–butadiene is least stable :–

(1) $\overset{\ominus}{C}H_2–CH=CH–CH=\overset{\oplus}{O}–CH_3$

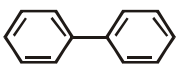
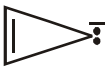
(2) $CH_2=CH–\overset{\ominus}{C}H–CH=\overset{\oplus}{O}–CH_3$

(3) $\overset{\ominus}{C}H_2–\overset{\oplus}{C}H–CH=CH–O–CH_3$

(4) $CH_2=CH–CH=CH–O–CH_3$

GC0046

44. Four structures are given in options (a) to (d). Examine them and select the aromatic strucutures.

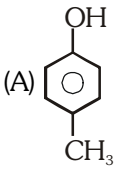
(a) 
(b) 
(c) 
(d) 

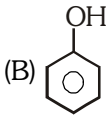
(1) a and d
(2) b and c
(3) a and b
(4) a and c

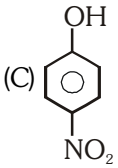
GC0047

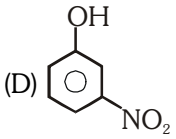
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45. Order of acidic strength of the following compound will be :

(A) 

(B) 

(C) 

(D) 

(1) C > D > B > A
(2) D > C > B > A
(3) A > B > C > D
(4) B > A > C > D

GC0048

46. Phenol is less acidic than
(1) Ethanol
(2) o-Nitrophenol
(3) o-Methylphenol
(4) o-Methoxyphenol

GC0049

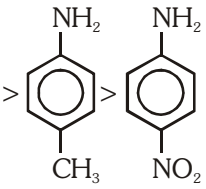
47. Temporary electron displacement effect in a molecule that occurs when a reagent approaches to attack if, is called as -
(1) Inductive effect
(2) Resonance effect
(3) Mesomeric effect
(4) Polarisability effect

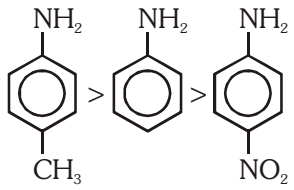
GC0164

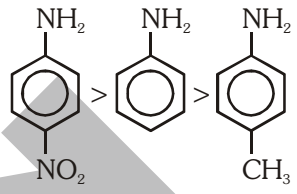
48. Which statement is incorrect.
(1) the energy of actual structure of the molecule is lower than that of any canonical structure
(2) The energy difference between actual structure and least energy resonance structure is called as resonance energy
(3) More number of resonating structure, more resonance
(4) In equivalent resonance structure of acetate ion of C=O bond length are unequal

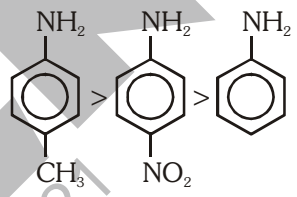
GC0165

49. Increasing order of basic strength is :-

(1) Ph – NH_2 > 

(2) 

(3) 

(4) 

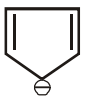
GC0166

50. Phenol are ortho para directing due to :
(1) —OH groups shows +M & –I
(2) —OH does not show hinderance
(3) The resonance effect increases the e^- density at o & p position
(4) The I-effect decreases the e^- density at meta position

GC0167

51. Which of the following is not aromatic ?

(1) 

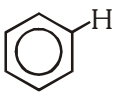
(2) 

(3) 

(4) 

GC0168

52. Arrange the following in their acidic strength order


(a)

$CH_3–CH_2–CH_2–H$ (b)

H–C≡C–H (c)

(1) a > b > c
(2) c > a > b
(3) b > a > c
(4) a > c > b

GC0169

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HYPERCONJUGATION

53. Which of the following compounds exhibits hyperconjugation :

(1) Phenol (2) Ethyne

(3) Ethanol (4) Propene

GC0051

54. Which of the following is least stable :–

(1) $CH_3-\overset{\oplus}{C}H-CH_3$ (2) $CH_3-CH_2-\overset{\oplus}{C}H_2$

(3) $CH_3-\underset{CH_3}{\overset{\oplus}{C}}-CH_3$ (4) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-\overset{\oplus}{C}H-C_6H_5$

GC0052

55. Which of the following is most stable alkene :-

(1) $H_2C=CH_2$

(2) $CH_3-CH=CH_2$

(3) $H_5C_2-CH=CH_2$

(4) $(CH_3)_2CH-CH=CH_2$

GC0053

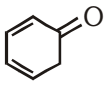
TAUTOMERISM

56. Which of the following will lead to maximum enolisation :–

(1) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

(2) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-H$

(3) $CH_3-\overset{O}{\overset{\|}{C}}-\underset{Br}{CH}-\overset{O}{\overset{\|}{C}}-CH_3$

(4) 

GC0056

57. Urea $\left[H_2N-\overset{O}{\overset{\|}{C}}-NH_2\right]$ molecule exhibits (isomerism):-

(1) Chain (2) Position

(3) Geometrical (4) Tautomerism

GC0057

58. Tautomerism is not observed in :–

(1) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (2) Ph–CH=CH–OH

(3) CH_3-NO_2 (4) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CHO$

GC0058

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	2	3	3	2	3	1	1	2	4	3	4	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	3	4	2	3	4	3	4	3	1	3	4	1	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	2	2	3	4	2	1	2	1	3	4	4	3	4	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58		
Ans.	2	4	4	2	3	4	2	4	2	2	4	4	4		

EXERCISE-III (Analytical Questions) Master Your Understanding

1. Consider the following statements :-

(a) $CH_3O\overset{\oplus}{C}H_2$ is more stable than $CH_3\overset{\oplus}{C}H_2$

(b) $Me_2\overset{\oplus}{C}H$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$

(c) $CH_2=CH-\overset{\oplus}{C}H_2$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$

(d) $CH_2=\overset{\oplus}{C}H$ is more stable than $CH_3\overset{\oplus}{C}H_2$

of these statements :-

(1) a and b are correct
(2) c and d are correct
(3) a, b and c are correct
(4) b, c and d are correct

GC0108

2. Which of the following carbon has most acidic hydrogen.

$\underset{④}{CH_3}-\underset{③}{CH_2}-\underset{②}{CH_2}-\underset{①}{\overset{\overset{O}{\|}}{C}}-H$

(1) C_1–H (2) C_2–H
(3) C_3–H (4) C_4–H

GC0109

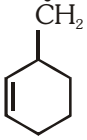
3. Hydrogen attached to sp^3 carbon in cyclopentadiene can be easily removed as what :-

(1) Hydride ion (2) Hydrogen molecule
(3) Proton (4) Hydrogen atom

GC0116

4. Which one is most stable free radical :

(1) 

(2) 

(3) $CH_2=CH-\dot{C}H_2$ (4) $CH_3-\overset{\overset{CH_3}{|}}{\dot{C}}-CH_3$

GC0117

5. Which is wrong electromeric effect :

(1) 

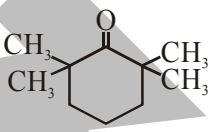
(2) $CH_3-CH=CH_2 \longrightarrow CH_3-\overset{\oplus}{C}H-\overset{\ominus}{C}H_2$

(3) $-C\equiv N \longrightarrow -\overset{\ominus}{C}=\overset{\oplus}{N}$

(4) $CH\equiv CH \longrightarrow \overset{\oplus}{C}H=\overset{\ominus}{C}H$

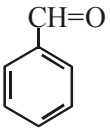
GC0118

6. Tautomerism is exhibited by :-

(1) 

(2) $CH_3-CH_2-N\overset{O}{\underset{O}{\rightleftharpoons}}$

(3) 

(4) 

GC0119

7. Identify the compound that exhibits tautomerism :-

(1) 2-Pentanone (2) Phenol
(3) 2-Butene (4) Lactic acid

GC0120

8. Arrange the following in correct order of acidic strength :

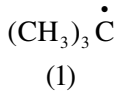
(I) $CH_3—NO_2$
(II) $NO_2—CH_2—NO_2$
(III) $CH_3—CH_2—NO_2$
(IV) $NO_2—\underset{\underset{NO_2}{|}}{CH}—NO_2$

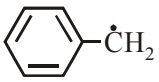
(1) IV > II > I > III
(2) IV > II > III > I
(3) III > I > II > IV
(4) III > I > IV > II

GC0121

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9. The correct order of stability of following carbon free radical is :-


(1)


(2)


(3)


(4)

(1) 1 > 2 > 3 > 4
(2) 4 > 3 > 2 > 1
(3) 2 > 3 > 1 > 4
(4) 2 > 1 > 3 > 4

GC0122

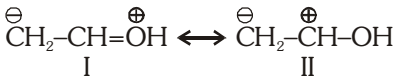
10. Which of the following is the strongest acid :-
(1) Carbolic acid
(2) Carbonic acid
(3) Picric acid
(4) Acetic acid

GC0123

11. The correct order of acidic strength is :-
(1) $CH_3COOH < HCN < H_2O < C_2H_5OH$
(2) $CH_3COOH > HCN > H_2O > C_2H_5OH$
(3) $HCN > CH_3COOH > H_2O > C_2H_5OH$
(4) $CH_3COOH > HCN > C_2H_5OH > H_2O$

GC0125

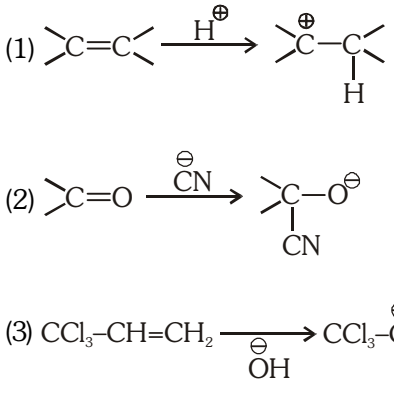
12. The two contributing structure of vinyl alcohol are:



Correct statement is/are :
(1) I is more stable than II due to complete octet
(2) II is more stable than I due to less covalent bond
(3) Both are less stable than vinyl alcohol due to more covalent bond
(4) 1 and 3 both

GC0173

13. Which is an example of +E effect ?

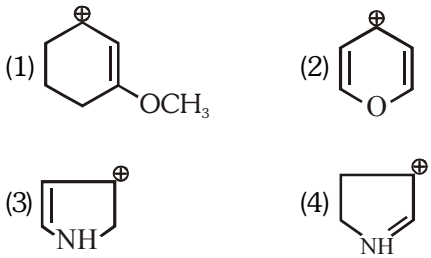


GC0174

14. Hyper conjugation involves delocalisation of except?
(1) Electrons of C–H bond towards directly attached unsaturated system
(2) Electrons of C–H bond towards directly attached positively charged carbon
(3) Electrons of C–H bond towards directly attached having unpaired electron
(4) Electrons of C–H bond towards directly attached negatively charged carbon

GC0175

15. Which cation have no resonance with lone pair



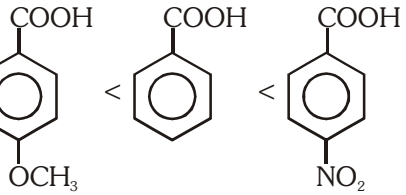
GC0176

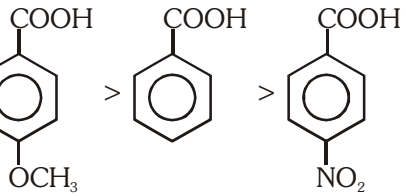
16. Which bond is more polar in following pair
(a) $CH_3–Br > CH_3–H$
(b) $CH_3–NH_2 < CH_3–OH$
(c) $CH_3–OH < CH_3–SH$
(1) a, c only (2) a, b, c
(3) a, b only (4) b,c only

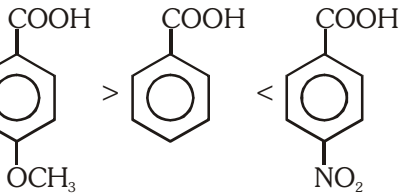
GC0177

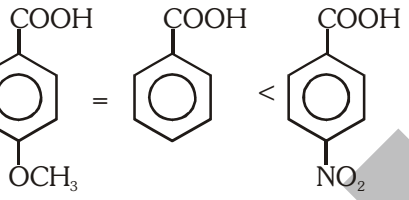
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17. Arrange the following derivative of carboxylic acid in their increasing order of their acidity.

(1) 

(2) 

(3) 

(4) 

GC0178

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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**2**	**3**	**1**	**3**	**2**	**1,2**	**1**	**4**	**3**	**2**	**4**	**1**	**4**	**4**
Que.	16	17													
Ans.	**3**	**1**													